Crop Recommendation System Using Random Forest Algorithm

Abstract

The Crop Recommendation System predicts the most suitable crops using soil and environmental parameters. The system uses the Random Forest machine learning algorithm to analyze factors such as nitrogen, phosphorus, potassium, pH, temperature, humidity, and rainfall to generate accurate crop recommendations.

Introduction

Agriculture increasingly relies on data-driven systems to improve productivity. Farmers often select crops based on experience rather than environmental data analysis. Machine learning models can analyze soil and climate factors to recommend optimal crops.

Problem Statement

Selecting crops based on environmental factors is complex and often inaccurate when relying only on experience. A machine learning-based system is needed to improve crop selection accuracy.

Methodology

The dataset is preprocessed to handle missing values and normalize features. The Random Forest algorithm is trained using environmental parameters to predict suitable crops. The trained model is used for predictions.

Dataset Features

The dataset includes Nitrogen, Phosphorus, Potassium, pH, Temperature, Humidity, and Rainfall.

Model Training and Evaluation

The dataset is split into training and testing sets. The Random Forest model builds multiple decision trees and aggregates predictions using a voting mechanism for accurate results.

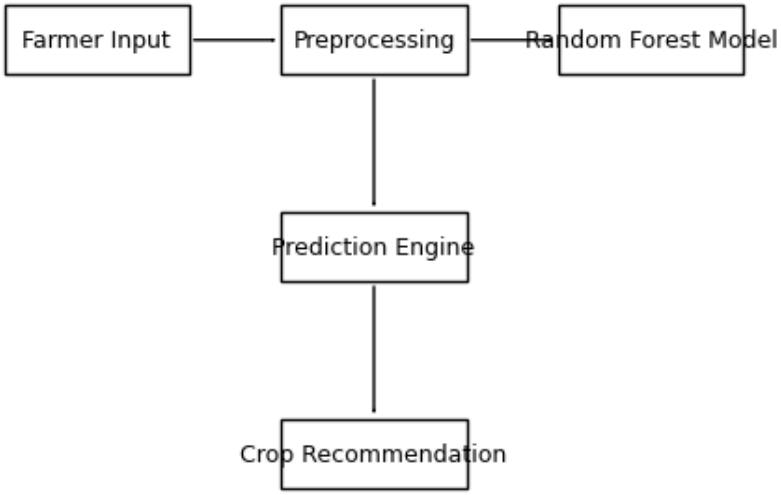
Results

The system provides accurate crop predictions that help farmers select crops suited to soil and climate conditions.

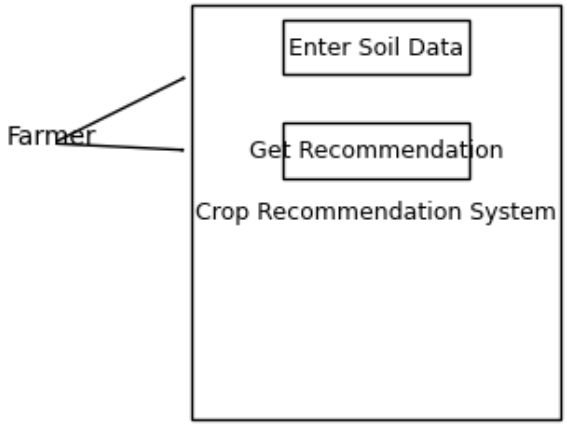
Conclusion

The proposed system demonstrates the effectiveness of machine learning in agriculture. Random Forest improves prediction accuracy and supports data-driven farming decisions.

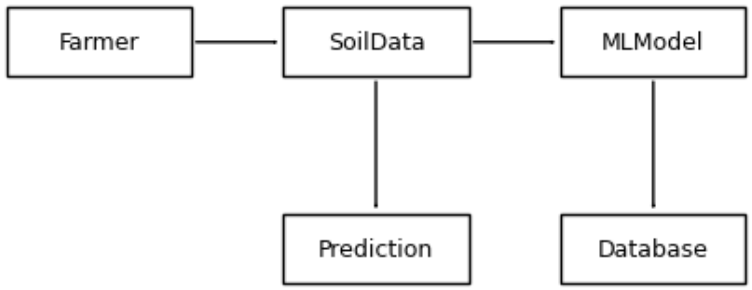
System Architecture Diagram



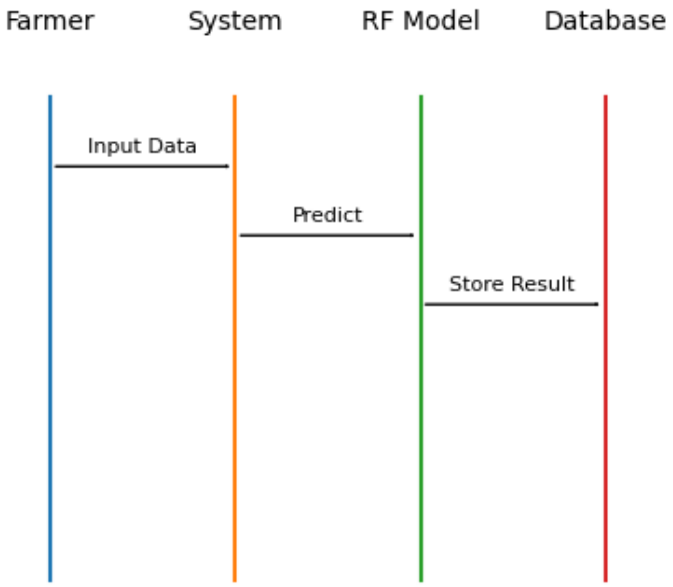
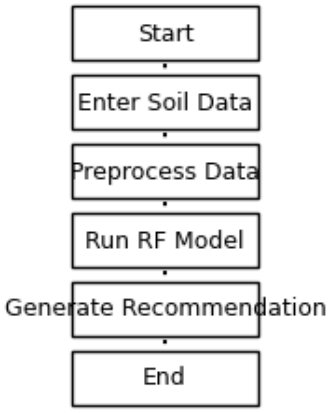
Use Case Diagram



Class Diagram



Sequence Diagram



Activity Diagram